3 Eliciting Requirements

A core activity of requirements engineering is the elicitation of requirements for the system to be developed. The basis for requirements elicitation is the knowledge that has been gained during requirements engineering about the system context of the system to be developed, which comprises the requirements sources that are to be analyzed and queried.

3.1 Requirements Sources

Three types of requirements sources

There are three different kinds of requirements sources:

- *Stakeholders* (see section 1.1.2) are people or organizations that (directly or indirectly) influence the requirements of a system. Examples of stakeholders are users of the system, operators of the system, developers, architects, customers, and testers.
- *Documents* often contain important information that can provide requirements. Examples of documents are universal documents, such as standards and legal documents, as well as domain- or organization-specific documents, such as requirements documents and error reports of legacy systems.
- *Systems in operation* can be legacy or predecessor systems as well as competing systems. By giving the stakeholders a chance to try the system out, they can gain an impression of the current system and can request extensions or changes based on their impressions.

3.1.1 Stakeholders and Their Significance

Significance of stakeholders

Identifying the relevant stakeholders is a central task of requirements engineering [Glinz and Wieringa 2007]. For the requirements engineer, stakeholders are important sources of requirements for the system (see section 1.1.2). It is the task of the requirements engineer to gather, document, and consolidate the partially conflicting goals and requirements of different stakeholders [Potts et al. 1994] (see chapter 8).

Consequences of unconsidered stakeholders

If stakeholders are not identified or not considered, it may result in significant negative repercussions for the project progress because requirements may remain undetected. At the latest, these overlooked requirements will enter the picture in the form of change requests during system operation. Fixing these issues retroactively causes high additional costs. Therefore, it is essential to identify all stakeholders and integrate them into the elicitation procedures.

Stakeholder lists provide overview.

An auxiliary technique for stakeholder identification is maintaining checklists. This allows for systematic and targeted elicitation of relevant stakeholders. If the stakeholder list is updated too late or incompletely, the result may be that important aspects of the system remain undetected, that the project goal is missed, or that significant additional costs arise from fixing issues. The starting point for stakeholder elicitation is often suggestions of relevant stakeholders that are made by management or by domain experts, for example. On the basis of these suggestions, relevant stakeholders can be identified.

3.1.2 Handling Stakeholders in the Project

Managing stakeholders

It can often be observed in practice that a lot of stakeholders are involved in complex and “difficult” projects. Due to limited resources, the stakeholders that

are the most suitable for requirements elicitation must be carefully selected. To document the stakeholders in the development process, it makes sense to use tables and spreadsheets that contain (at least) the following data: name, function (role), additional personal and contact data, temporal and spatial availability during the project progress, relevance of the stakeholder, area and extent of expertise of the stakeholder, and the stakeholder's goals and interests regarding the project.

Making collaborators out of the affected

Handling stakeholders also means continuously exchanging information: Periodic status updates and continuous involvement of the stakeholders assist the requirements engineer in turning people previously simply affected by the project (i.e., principally affected stakeholders) into collaborators (i.e., well-integrated, jointly responsible stakeholders).

Individual "contracts" with the stakeholders

Stakeholders that are not given enough attention by the requirements engineer might be overly critical toward the project. In addition, some stakeholders may show a lack of motivation because they are sufficiently satisfied with the legacy system, are afraid of change, or are negatively biased due to previous projects. It's the requirements engineer's task to support the project manager in convincing all stakeholders of the benefit of the project. To avoid misunderstandings and disputes regarding competence, it is useful to formally agree on the tasks, responsibilities, and managerial authority as well as to determine individual goals, communication paths, and feedback loops that can be used by the stakeholders. Depending on the culture of the organization, this agreement and determination can be done verbally (i.e., by "shaking hands") or, more formally, by means of written documentation. The individual agreements should be signed off by the managers.

Obligations and privileges of the stakeholders

A number of obligations and privileges result from the agreement with the stakeholders.

The requirements engineer

- speaks the language of the stakeholders,
- becomes thoroughly familiar with the application domain,
- creates a requirements document,
- is able to get work results across (e.g., by means of diagrams and graphs),
- maintains a respectful relationship with any stakeholder,
- presents her ideas and alternatives as well as their realizations,
- allows stakeholders to demand properties that make the system user-friendly and simple,
- ensures that the system satisfies the functional and qualitative demands of the stakeholders.

The stakeholders

- introduce the requirements engineer to the application domain,
- supply the requirements engineer with requirements,
- document requirements assiduously,
- make timely decisions,
- respect the requirements engineer's estimates of costs and feasibility,
- prioritize requirements,
- inspect the requirements that the requirements engineer documents, such as prototypes, etc.,
- communicate changes in requirements immediately,
- adhere to the predetermined change process,
- respect the requirements engineering process that has been instated.

Elicitation techniques determine communication and process.

In addition, the requirements engineer plans and organizes the communication paths as well as drafts a structured schedule for the requirements engineering activities that are to be performed in collaboration with the stakeholders. This organization and the type of communication are significantly influenced by the elicitation techniques that can be used during requirements engineering.

3.2 Requirements Categorization According to the Kano Model

Influence of the requirements on satisfaction

Knowing the importance of a requirement for the satisfaction of the stakeholders is very helpful for requirements elicitation. Along with the respective properties of a product that determine the satisfaction, the satisfaction is classified into the following three categories [Kano et al. 1984]:

- *Dissatisfiers* are properties of the system that are self-evident and taken for granted (subconscious knowledge).
- *Satisfiers* are explicitly demanded system properties (conscious knowledge).
- *Delighters* are system properties that the stakeholder does not know or expect and discovers only while using the system—a pleasant and useful surprise (unconscious knowledge).

As time goes by, delighters turn into satisfiers and dissatisfiers as the user becomes accustomed to the properties of the system. When eliciting requirements, all three categories must be considered.

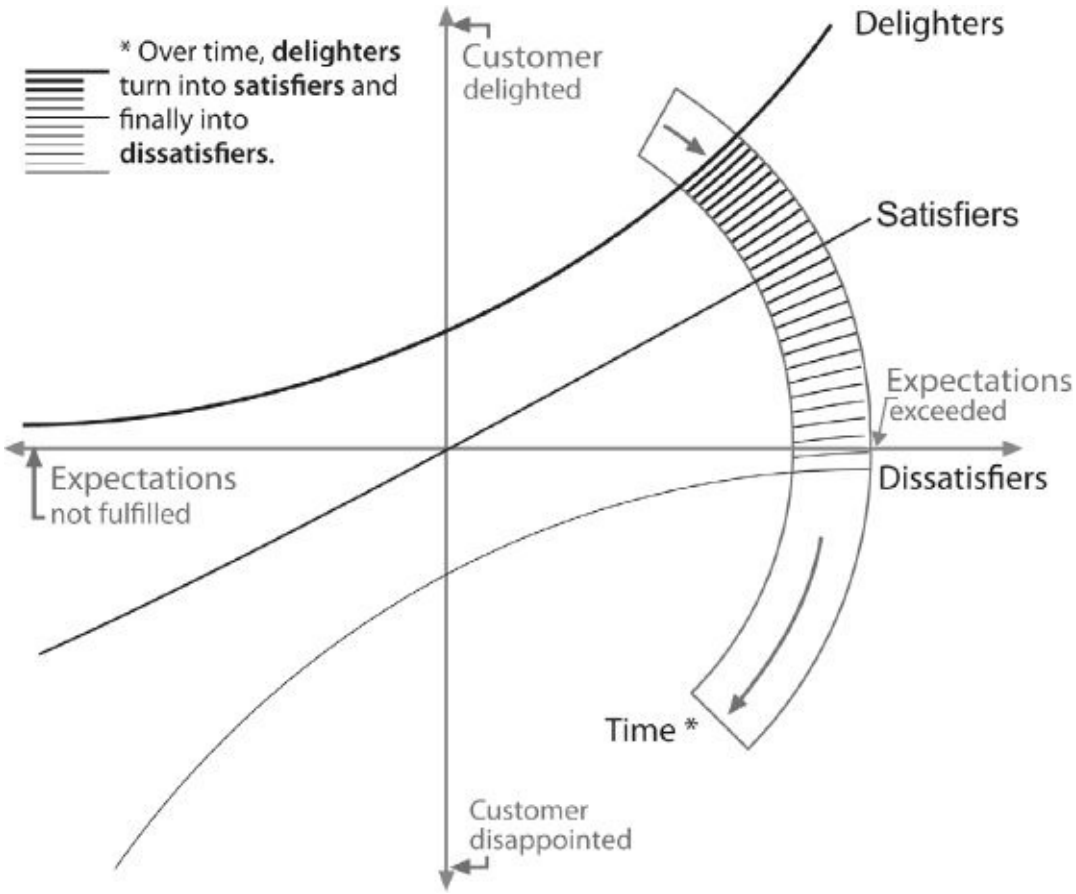


Figure 3-1 *Graphical representation of the Kano model*

Dissatisfiers

Dissatisfiers (subconscious requirements) must be fulfilled by the system in any case. Otherwise, stakeholders will be disappointed and dissatisfied. Completely fulfilled dissatisfiers do not generate a positive disposition but merely help to avoid massive discontent. Dissatisfiers are dominantly influenced by existing systems. Therefore, observation and document-centric techniques are especially well suited for the elicitation of these factors.

Satisfiers

Satisfiers (conscious requirements) are properties that are consciously known to the stakeholders and explicitly demanded. When these properties are fulfilled, stakeholders are content and satisfied, which is desirable. If some demanded properties are missing, the stakeholders probably will not accept the product. Their satisfaction decreases with each missing satisfier. Satisfiers can be elicited well using survey techniques.

Delighters

Delighters (unconscious requirements) are properties of a system whose value is recognized only when the stakeholder can try out the system for herself or the requirements engineer proposes them. Creativity techniques are best suited to elicit delighters.

3.3 Elicitation Techniques

Requirements elicitation: no universal method

The main goal of all elicitation techniques is in supporting the requirements engineer in ascertaining the knowledge and requirements of the stakeholders. How and when a technique can be applied depends on the given conditions. Applying the technique consciously and in a fashion appropriate to the situation at hand allows for tailoring the requirements elicitation process which takes into account project constraints so that requirements may be elicited as completely and

comprehensibly as possible.

3.3.1 Types of Elicitation Techniques

Influencing factors regarding the choice of elicitation techniques

Elicitation techniques serve the purpose of identifying the conscious, unconscious, and subconscious stakeholder requirements. However, there is no universal method to elicit these requirements [Hickey and Davis 2003]. Every project has individual constraints and individual characteristics and is by and large unique, but there are always elicitation techniques that are compatible with the project. The most important influencing factors when choosing the appropriate elicitation techniques are as follows:

- the distinction between conscious, unconscious, and subconscious requirements that are to be elicited
- the time and budget constraints, as well as the availability of the stakeholders
- the experience of the requirements engineer with a particular elicitation technique
- the chances and risks of the project

Risk factors

The first important step when choosing a suitable elicitation technique is to perform an analysis of constraints critical to the project, i.e., identifying so-called risk factors. Mostly, these result from human, organizational, and professional influences, as illustrated in the following passages.

Human influences

During the requirements elicitation phase, which is heavily influenced by the stakeholders, good communication is essential. In order to assure high-quality communication between the requirements engineer and stakeholders, it is important to determine the type of requirement, the desired level of detail, and the experience of the requirements engineer and the interviewees with different elicitation techniques.

Social, group-dynamic, and cognitive capabilities of the stakeholders also

influence the choice of suitable elicitation techniques significantly. Another influence factor is whether the elicited knowledge is explicit (consciously known) by each individual stakeholder or if it is implicit or unconscious (i.e., covert).

Organizational influences

Organizational risk factors the project faces need to be investigated as well. Among other things, this comprises the distinction between fixed price contracts and service contracts, whether the system to be built is a new development or an extension of a legacy system, and spatial and temporal availability of the stakeholders.

Operational influences of the content

In addition, it is necessary to consider the operational content of the requirements. If the system is very complex, it is advisable to employ a structuring approach during elicitation in order to deconstruct the operational contents into understandable parts.

Combine techniques with regard to your particular situation to lower risks.

Another influencing factor on the choice of elicitation techniques is the desired level of detail of the requirements. Abstract requirements can be elicited rather well using creativity techniques. With the stakeholders, a vision of the system or its important properties can be created or collected. Inquisitive (survey) techniques or observational techniques can aid in eliciting requirements of a medium level of detail [Robertson 2002]. Finely detailed requirements can be elicited well by making use of document-centric techniques, i.e., techniques that use existing documents because information up to an arbitrary level of detail can be extracted from these.

It is advisable to combine different techniques because this minimizes many of the risks inherent to the project. Weaknesses and pitfalls of a particular technique can be balanced out through the use of another technique whose strong points lie where the first technique may have deficits.

3.3.2 Survey Techniques

Eliciting explicit knowledge

Survey techniques aim at eliciting as precise and unbiased statements as possible from stakeholders regarding their requirements. All survey techniques assume that the respondent is capable of explicitly expressing his or her knowledge and that he or she is committed to investing time and effort for the elicitation. Survey techniques are usually driven by the requirements engineer because she asks the questions. This, however, might result in the fact that stakeholder concerns are forgotten, superseded, or disregarded.

Interview

- During an *interview*, the requirements engineer asks predetermined questions to one or more stakeholders and documents the answers. Questions that arise during the conversation can be discussed immediately, and the requirements engineer may uncover subconscious requirements through clever questions. Interviews can be employed during the entire development phase of the system. An experienced interviewer individually controls the course of the conversation, completely commits herself to each stakeholder, inquires about specific aspects, and thus ensures the completeness of the answers. The most prominent disadvantage of this elicitation technique is that it is very time-consuming.

Questionnaire

- *Questionnaire:* Making use of open and/or closed questions (e.g., multiple choice questions) is another way of eliciting requirements from stakeholders. If there are a large number of participants that must be surveyed, an online questionnaire is a viable option. Questionnaires can elicit a magnitude of information in a short amount of time and at low costs. As long as answers are predetermined, even stakeholders that are not able to explicitly express their knowledge can deliver an assessment. A disadvantage of using a questionnaire is that it can be only employed to gather requirements the requirements engineer already knows or conjectures. Creating a proper questionnaire is often tricky and time-consuming and requires thorough knowledge of the domain in question and the psychological guidelines for creating questionnaires. In addition, as opposed to interviews, questionnaires do not provide immediate feedback between the surveyor and the surveyed,

so it becomes apparent that questions were forgotten or badly formulated only once the questionnaires have been evaluated.

3.3.3 Creativity Techniques

Establishing innovations

Creativity techniques serve the purpose of developing innovative requirements, delineating an initial vision of the system, and eliciting excitement factors. Creativity techniques are usually not well suited for establishing fine-grained requirements about the system behavior. The following creativity techniques are commonly used [Maiden and Gizikis 2001]:

Brainstorming

- During *brainstorming*, ideas are collected within a certain time frame, usually in groups of 5 to 10 people. The ideas are documented by a moderator without discussing, judging, or commenting on them at first. Participants use ideas of other participants to develop new original ideas or to modify existing ideas. After that, the collected ideas are subjected to a thorough analysis. This technique is especially effective when a large number of people of different stakeholder groups are involved. Among the advantages of this technique is that a large number of ideas can be collected in a short amount of time and multiple people can expand on these ideas collaboratively. The unbiased collection of these ideas allows new solutions to pop up. Brainstorming is usually less effective when the dynamics of the group are muddled or when participants with very varied levels of dominance are involved. For such situations, other creativity techniques may be better suited, e.g., the 6-3-5 method (six participants, three ideas each, fivefold hand-off of the ideas) [Rohrbach 1969] or the brainwriting method.

Brainstorming paradox

- *Brainstorming paradox* is a modification of regular brainstorming in that events that must not occur are collected. Afterward, the group develops measures to prevent the events collected earlier from happening. Through this process, participants often realize which actions may entail negative

results. With this method, risks can be identified early on and countermeasures can be developed. Advantages and disadvantages of this technique are identical to those of classic brainstorming.

Change of perspective

- *Change of perspective:* Among the techniques that employ a change of perspective (adopting different extreme standpoints), the most common technique is the so-called Six Thinking Hats [DeBono 2006]. Each of the six hats represents a particular perspective that is in turn adopted by each of the participants. The resulting solutions approach the problem from different standpoints. That way, even stakeholders that are very convinced of their own opinion are persuaded to adopt a different standpoint. This technique is extraordinarily beneficial when stakeholders can only express their knowledge in a biased manner or are harshly constricted to their opinions. On the other hand, this technique cannot be applied if the requirements require a fine-grained level of detail because this would render the technique very laborious.

Analogy technique

- *Analogy techniques (bionics/bisociations):* In bionics, problems that the project faces are mapped to an analogous situation occurring in nature, and the solutions nature provides are sought and then mapped back to the project. In bisociation, the analogies need not originate in nature. These techniques assume that each participant is capable of analogous thinking, that a lot of time is available, and that the participants have an in-depth knowledge of the domain with which an analogy will be drawn. Analogy techniques can be applied covertly or in the open. When this technique is applied covertly, the participants are only told the analogy. The requirements engineer is then responsible for mapping the results onto the real problem space. When this technique is applied in the open, the stakeholders know the real problem space as well as the analogy.

3.3.4 Document-centric Techniques

Document-centric techniques reuse solutions and experiences made with existing

systems. When a legacy system is replaced, this technique ensures that the entire functionality of the legacy system can be identified. Document-centric techniques should be combined with other elicitation techniques so that the validity of the elicited requirements can be determined and new requirements for the new system can be identified.

System archaeology

- *System archaeology* is a technique that extracts information required to build a new system from the documentation or implementation (code) of a legacy system or a competitor's system. The technique is often applied when explicit knowledge about the system logic has been lost partially or entirely. By analyzing existing code, the requirements engineer ensures that none of the functionalities of the legacy system will be overlooked and the system logic of the legacy system is elicited anew. This method leads to a large amount of very detailed requirements and is very laborious. However, system archaeology is the only technique that can ensure that all functionalities of the legacy system will be implemented in the new system. When it becomes obviously apparent that the legacy system and the new system differ in functionality, additional elicitation techniques, e.g., creativity techniques, must be applied early on.

Perspective-based reading

- *Perspective-based reading* (see section 7.5.4) is applied when documents need to be read with a particular perspective in mind, e.g., the perspective of the implementer or the tester. Aspects that are contained in the document but do not pertain to the current perspective are ignored. This allows for an analysis that is strictly focused on particular parts of the existing documentation. This way, detailed, technology-related or implementation-related aspects can be separated from essential operational aspects that are relevant for the successor system.

Reuse

- *Reuse:* Requirements that have been previously compiled and brought up to a certain quality standard can be reused. In order to do that, the requirements are stored in a database, for instance, and kept available at the required level

of detail for reuse. Through reuse, the costs involved with the elicitation procedures can be significantly reduced.

3.3.5 Observation Techniques

Question observations and optimize processes.

When domain specialists are unable to spend the time needed to share their expertise with the requirements engineer, or are unable to express and denote their knowledge, observation techniques are helpful. During observation, the requirements engineer observes the stakeholders while they go about their work. The requirements engineer documents all steps and thus elicits the processes the system must support as well as potential mistakes, risks, and open questions. All those are potential requirements that need to be formulated. The stakeholders can actively demonstrate their knowledge in using the application or can remain passive, with the requirements engineer merely observing. The requirements engineer ought to question the observed processes so that the situation as it should be can be established. Otherwise, she is at risk of documenting outdated technological decisions and suboptimal processes (i.e., the situation as is and not as it should be). As the requirements engineer is an external observer, her chances of identifying inefficient processes are good and she can then suggest better solutions. She is farther removed from the processes than the stakeholders, who frequently repeat work steps without questioning them critically. Observation techniques are well suited to elicit detailed requirements and dissatisfiers because the requirements engineer can recognize dissatisfiers thought of as self-evident or only subconsciously known by the stakeholders. In addition, the requirements engineer becomes very familiar with the domain language, which simplifies further elicitation. Satisfiers can only be observed if they have been implemented in the legacy system or are actively employed in the current processes. As a result, this technique is not suited for the development of new processes. During system development, field observations and apprenticing are especially well suited as elicitation techniques.

Field observation

- *Field observation:* The requirements engineer is on location with the specialist or the users of the system and observes and documents the

processes and operational procedures that they carry out. Using these observations, she formulates the requirements. Often, this can be further aided by audio and video recordings. This technique is well suited for operational procedures that are difficult to express verbally, but it can only be applied if the procedures are visible physically.

Apprenticing

- With *apprenticing*, the requirements engineer must actively learn and perform the procedures of the stakeholders. Just like an apprentice, the requirements engineer is encouraged to question unclear and complex operational procedures so that she may gather domain experience. Thereby, she can experience requirements that the stakeholders take for granted and therefore cannot elucidate. Another advantage is that the typical balance of power between the requirements engineer and the respective specialist is reversed because the stakeholder now adopts the role of the "master" that has the knowledge the apprentice is yet lacking.

3.3.6 Support Techniques

Support techniques serve as an addition to the elicitation techniques and try to balance out the weaknesses and pitfalls of the chosen elicitation technique.

Mind mapping

- In *mind mapping*, a graphical representation of the refined relationships and interdependencies between terms is created. Mind mapping is often used as a supporting technique for brainstorming or brainstorming paradox.

Workshops

- During a joint meeting, the requirements engineer and the stakeholders elaborate the goals (or details of a certain functionality) of the system. For example, the necessary user interfaces of the system can be designed in a *workshop* [Gottesdiener 2002].

CRC cards

- With the CRC technique (CRC stands for *Class Responsibility Collaboration*), context aspects and their respective attributes and properties are denoted on index cards. Requirements are then formulated using these cards.

Audio and video recordings

- *Audio and video recordings* are very well suited to elicit requirements when stakeholders are not always available, when budget is tight, or when the system is highly critical. Especially during field observations, audio and video recordings can help capture fast-paced processes. The disadvantage of this technique is that stakeholders often feel supervised when they are being recorded and as a result might deliver biased statements or, in extreme cases, might even refuse to cooperate.

Modeling action sequences

- *Use case modeling:* Use cases document the external view of the system to be developed. A use case has a trigger event, which triggers the use case and an expected result, or outcome of the use case. Every use case is a functionality that must be supported by the system to be developed (see section 6.3).

Prototypes for illustration

- *Prototypes* are well suited to question established requirements and to elicit requirements in situations where stakeholders have only a vague understanding of what is to be developed. Potential consequences of new or changed requirements can be identified easier. For example, user interface prototypes are frequently used in practice to find additional functional requirements.

3.4 Summary

Requirements elicitation is a core activity in requirements engineering. Aside from documents and legacy systems, stakeholders are the main sources for requirements. It is important to initially agree upon mutual rights and

responsibilities of the stakeholders and the requirements engineer in order to facilitate communication and cooperation and to successfully integrate the stakeholders into the elicitation process. The choice of the right elicitation technique for the respective project is made by the requirements engineer based on the given cultural, organizational, and domain-specific constraints.